Collapse / Resolution Framework (CRF)

A Computational Framework for Born-Aligned Selection, Post-Selection Dynamics, and Constraint Stabilization

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Working Paper Notice

This document presents computational and structural investigations into post-selection dynamics within a constrained toy architecture referred to as Collapse / Resolution Framework (CRF).

CRF is an experimental simulation framework designed to study how Born-aligned first selection behaves under asymmetric detector geometry, dwell-dependent survival dynamics, and stabilization constraints. The framework does not claim violation of quantum mechanics, replacement of the Born rule, or discovery of new physical laws. All results are bounded to the tested architecture and are presented as computational findings rather than physical proofs.

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1. Abstract

We report a series of computational validation experiments (§26–§41) on a three-component simplex dynamical system (omniflow: L+S+Q+D with static bias field H) as a test bed for Collapse / Resolution Framework (CRF). The central question is whether a layered selection architecture — Born-weighted first selection, followed by exposure dwell, followed by a Survival / Capture operator, followed by Born-fallback resolution — can reproduce Born-consistent statistics under ideal conditions while producing systematic deviations under asymmetric detector geometry.

We find that:

1. First selection via an exponential race faithfully encodes Born probabilities p for all tested regimes.

2. The Survival / Capture layer

$$s_i = 1 - e^{-\kappa\tau_i}$$

acts as a post-selection filter that can amplify, preserve, or partially invert Born weighting depending on (κ,decay) geometry. 3. Born-fallback resolution (Fix B) eliminates all unresolved outcomes without modifying the survival formula. 4. Born reduction — final outcomes converging back to p — occurs reliably only under symmetric detector geometry or high-κ lock-in.

Six representative detector types are placed on the parameter map; three fall in the Born-like region (PD, APD, SNSPD) and two show strong deviation (Trapped Ion Fluorescence, Superconducting Qubit Readout).

Sections §37–§41 extend the architecture to include a dwell-asymmetry stabilizer with coupling parameter γ, identifying:

$$\gamma \approx 0.75$$

as the first admissible value at coarse resolution and a robust interior operating point within a constrained admissible window.

The survival layer is found to be irreducibly full-configuration dependent.

The architecture is now complete enough to write honestly.

All results are experimental. No quantum-mechanical claims are made.

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2. Introduction

CRF proposes that the observed Born distribution emerges from a multi-layer process rather than a single rule.

The central architectural claim is:

> Born-weighted selection at the first-selection level is necessary but not sufficient to guarantee Born-consistent final statistics.

A post-selection Survival / Capture layer can reshape outcomes, and the degree of reshaping depends on detector geometry, not on the quantum state itself.

The omniflow model:

$$L + S + Q + D$$

on the 3-simplex with static bias field H, provides the dynamical context for the CRF validation program.

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3. Core CRF Architecture

CRF is organized into four sequential layers.

Layer 1 — Born Selection

Each state draws a race time:

$$T_i \sim \text{Exp}(\Lambda p_i)$$

Winner:

$$\text{first} = \arg\min_i T_i$$

Because the exponential distribution is memoryless:

$$P(\text{first} = i) = p_i$$

exactly.

Layer 2 — Exposure / Dwell

Detector signal:

$$x_i(t) = A_i e^{-(t - T_i)/\text{decay}_i}$$

with Gaussian noise clipped to nonnegative values.

Threshold dwell:

$$\tau_i = \int 1[x_i(t) > \theta]dt$$

This converts race outcomes into kinetic exposure.

Layer 3 — Survival / Capture

The central post-Born layer:

$$s_i = 1 - e^{-\kappa\tau_i}$$

Longer dwell implies higher survival probability.

Key empirical result:

- Symmetric detector geometry preserves Born alignment.
- Asymmetric geometry amplifies whichever state dwells longest.

The Born rule governs first selection.

The survival layer governs whether that selection persists.

Layer 4 — Resolution / Born Fallback

If the selected channel survives:

$$u < s_{\text{first}}$$

then:

$$\text{final} = \text{first}$$

Otherwise reroute weights are formed:

$$\text{weight}_j = p_j s_j$$

and sampled categorically.

Fallback condition:

$$\sum_j \text{weight}_j = 0$$

triggers Born fallback resolution.

3.6 Operator Architecture

The PBST pipeline separates into four operators:

$$\mathcal{D}: (p, \theta) \to x$$

competition geometry.

$$\tau: x \to \tau$$

above-threshold dwell.

$$s: \tau \to s$$

raw survival:

$$s_i = 1 - e^{-\kappa \tau_i}$$

Stabilized survival:

$$\mathcal{P}_\gamma : s \to s^*$$

with:

$$s_i^* = \bar{s} + \lambda(s_i - \bar{s})$$

and:

$$\lambda = \frac{1}{1 + \gamma A_\tau}$$

Full pipeline:

$$p, \theta \to [\mathcal{D}] \to x \to [\tau] \to \tau \to [s] \to s \to \left[\mathcal{P}_\gamma\right] \to s^* \to \text{Resolution} \to \text{outcome}$$

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4. Mathematical Formulation

4.1 Exponential Race

$$T_i \sim \text{Exp}(\Lambda p_i)$$

$$P(\text{first} = i) = p_i$$

4.2 Detector Signal and Dwell

$$x_i(t) = A_i e^{-\max(t - T_i, 0)/\text{decay}_i} 1[t \geq T_i] + \varepsilon_i(t)$$

Noise:

$$\varepsilon_i(t) \sim N(0, \sigma^2)$$

Dwell:

$$\tau_i = \int_0^{T_{\text{total}}} 1\,[x_i(t) > \theta] dt$$

4.3 Survival / Capture

$$s_i = 1 - e^{-\kappa \tau_i}$$

with:

$$s_i \in [0,1]$$

4.4 Reroute Rule

$$\text{weight}_j = p_j s_j$$

Conditional reroute probability:

$$P(\text{final} = j \mid \text{fail}) = \frac{\text{weight}_j}{\sum_k \text{weight}_k}$$

4.5 Born Fallback

If:

$$\sum_j \text{weight}_j = 0$$

then fallback samples from:

$$P(j) \propto p_j$$

7. Parameter Map (§35)

36 detector configurations were swept:

$$\kappa \in \{0.3, 1.0, 3.0\}$$

$$\theta \in \{0.15, 0.25, 0.35\}$$

Decay asymmetry emerges as the dominant axis.

Key finding:

- Symmetric decay preserves Born-like behavior.
- Moderate asymmetric decay produces strongest deviation.
- High-κ lock-in restores Born behavior.

8. Detector Placement (§36)

Representative placements:

Detector	Region	L1
PD	Born-like	0.0135
APD	Born-like	0.0233
SNSPD	Born-like	0.0065
PMT	Moderate	0.0998
TIF	Strong deviation	0.2503
SQR	Strong deviation	0.2201

Structural interpretation:

- Near-symmetric decay + high κ naturally restore Born-like behavior.
- Strong asymmetry + moderate κ produce large deviation.

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9. Empirical Tension / Missing Stabilizer

The core tension:

> The Survival / Capture layer has no internal calibration mechanism.

The architecture predicts systematic deviation under asymmetric detector geometry unless:

1. Real detectors are physically calibrated,
2. A missing stabilizer exists,
3. Or the toy detector model is wrong.

No resolution is claimed.

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12. Conclusion

The exponential race selector is the first genuine multichannel Born-aligned selector found in this simulation program.

The Survival / Capture layer:

$$s_i = 1 - e^{-\kappa \tau_i}$$

acts as the central post-Born filter.

Born reduction occurs reliably only under:

1. Symmetric detector geometry,
2. Or very high-κ lock-in.

The open question remains:

> Are real detectors calibrated into Born-like regions, or is a missing stabilizer required?

All results are experimental.

No quantum-mechanical claims are made.

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§37. Dwell-Asymmetry Stabilizer

Raw Survival

$$s_i = 1 - e^{-\kappa \tau_i}$$

Stabilized Survival

Born-weighted mean survival:

$$\bar{s} = \sum_j p_j \, s_j$$

Dwell asymmetry:

$$A_\tau = 1 - \frac{\min_i(\tau_i + \varepsilon)}{\max_i(\tau_i + \varepsilon)}$$

Compression factor:

$$\lambda = \frac{1}{1 + \gamma A_\tau}$$

Stabilized survival:

$$s_i^* = \bar{s} + \lambda(s_i - \bar{s})$$

Properties:

$$\gamma = 0 \Rightarrow s_i^* = s_i$$

$$A_\tau = 0 \Rightarrow s_i^* = s_i$$

$$\gamma \to \infty \Rightarrow s_i^* \to \bar{s}$$

Recommended:

$$\gamma \approx 1.0$$

Interpretation:

The stabilizer suppresses runaway dwell asymmetry while preserving overall CRF structure.

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§38. Gamma Selection Validation

Empirical admissible window:

$$\gamma_{\min} \in (0.63, 0.70]$$

Working value:

$$\gamma_{\text{work}} \approx 0.75$$

Selection criterion:

- TIF and SQR reduced below threshold,
- Born-like detectors remain Born-like,
- No collapse to uniform.

Conclusion:

$$\gamma \approx 0.75$$

is not a lucky banana.

Purpose

Compare:

1. Raw survival dynamics:

$$\gamma = 0$$

2. Stabilized dynamics:

$$\gamma = 0.75$$

using identical detector suites and metrics.

Results

Detector	L1 raw	L1 stab	ΔL1
PD	0.0135	0.0257	−0.0122
APD	0.0219	0.0223	−0.0005
SNSPD	0.0059	0.0297	−0.0238
TIF	0.2424	0.1245	+0.1179
SQR	0.2228	0.1195	+0.1033
s35_worst	0.3577	0.1723	+0.1854
EXT_AS	0.2143	0.1130	+0.1013
EXT_PX	0.3923	0.1862	+0.2060

Interpretation:

- Born-like detectors naturally remain Born-like without stabilization.
- Asymmetric detectors generate strong deviation.
- Stabilization suppresses but does not erase deviation.

The stabilizer behaves as an admissibility constraint rather than a universal correction.

Final structural interpretation:

detector geometry → dwell asymmetry → survival amplification → frequency drift

CRF therefore closes not as a Born-breaking theory, but as a Born-preserving constraint architecture.

Closing Statement

All results remain experimental.

No core CRF formulas were modified.

All numerical outputs are reproducible with:

$$\text{seed} = 42$$

The framework remains a constrained toy architecture for studying post-selection dynamics, detector asymmetry, and admissibility structure.

I created the verbatim-style rendered document with the CRF structure and properly typeset math from your provided text, including the stabilizer, operator architecture, and §44 ablation material.

CRF ablation new